

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE**Supplementary Winter Examination – 2024****Course: B. Tech. Branch : Electronics & Computer Engg/Electronics & Computer Science Engg****Subject Code & Name: BTECPC602 - Artificial Intelligence and Machine Learning Semester :VI****Max Marks: 60****Date:20/12/2024****Duration: 3 Hr.****Instructions to the Students:**

1. Each question carries 12 marks.
2. Question No. 1 will be compulsory and include objective-type questions.
3. Candidates are required to attempt any four questions from Question No. 2 to Question No. 6.
4. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question.
5. Use of non-programmable scientific calculators is allowed.
6. Assume suitable data wherever necessary and mention it clearly.

		(Level/CO)	Marks
Q. 1	Objective type questions. (Compulsory Question)		12
1	In supervised learning, the goal is to: a. Discover hidden patterns in data. b. Make predictions based on labeled data. c. Group similar items without labeled data. d. Optimize a reinforcement signal.	(Analyze)	1
2	What type of environment is considered "episodic"? a. The agent's current action does not depend on previous actions. b. The agent must consider future consequences for current actions. c. The environment provides all necessary information at every step. d. The agent's actions influence the environment's rules.	(Remember)	1
3	What is the primary purpose of the "Generate and Test" method in state space search? a. To generate the best solution directly b. To generate possible solutions and test their validity c. To eliminate all incorrect solutions before testing d. To minimize the number of states generated	(Understand)	1
4	Why is Depth-First Iterative Deepening often preferred over DFS in infinite or large state spaces? a. It finds the solution faster than DFS.	(Analyze)	1

	b. It uses more memory but guarantees an optimal solution.					
	c. It combines the space efficiency of DFS with the completeness of BFS.					
	d.It relies on heuristics for its operation.					
5	In Depth-First Search, which data structure is typically used to keep track of unexplored nodes?				(Analyze)	1
	a. Queue	b. Stack	c. Priority Queue	d. Hash Table		
6	Which search algorithm guarantees finding an optimal solution for an unweighted graph?				(Remember)	1
	a. DFS	b. BFS	c. Depth-Bounded DFS	d. Generate and Test		
7	Which of the following is an example of unsupervised learning?				(Understand)	1
	a. Linear regression	b. K-means clustering	c. Decision trees	d. Gradient boosting		
8	Which of the following is a commonly used metric for regression problems?				(Understand)	1
	a. Accuracy	b. Mean Squared Error (MSE)	c. Precision	d. Recall		
9	What does the confusion matrix represent?				(Remember)	1
	a. The relationship between input and output data					
	b. A table that shows actual vs. predicted classifications					
	c. The correlation between features in a dataset					
	d. The mean of true positive and true negative rates					
10	Which metric provides a balance between precision and recall?				(Analyze)	1
	a. Classification accuracy	b. AUC-ROC	c. F1 Score	d. Support		
11	What is the primary objective of linear regression?				(Understand)	1
	a. Minimize variance in predictions					
	b. Predict binary outcomes					
	c. Minimize the cost function					
	d. Maximize the correlation coefficient					
12	The sigmoid function used in logistic regression outputs values between:				(Analyze)	1
	a. -1 and 1	b. 0 and 1	c. 0 and ∞	d. $-\infty$ and 0		

Q.2	Solve the following.		12
A)	Define: Artificial Intelligence. State what is need of AI?	(Remember)	6
B)	What is an agent in AI? What are the types of agent program in any intelligent system?	(Understand)	6
Q.3	Solve the following.		12
A)	Sate and explain Heuristic Search with example.	(Remember)	6
B)	What is a TABU Search? Explain in detail.	(Understand)	6
Q.4	Solve Any Two of the following.		12
A)	How does supervised learning differ from unsupervised learning ?	(Understand)	6
B)	Define inductive bias and explain its importance in machine learning.	(Analyze)	6
C)	What is cross-validation , and why is it used in machine learning? What are the different types of cross validation explain in detail.	(Remember)	6
Q.5	Solve Any Two of the following.		12
A)	How can you calculate precision, recall, and accuracy using a confusion matrix?	(Analyze)	6
B)	Differentiate between classification and regression in ML.	(Understand)	6
C)	Compare the utility of AUC-ROC and F1 score for evaluating a binary classifier.	(Evaluation)	6
Q.6	Solve Any Two of the following.		12
A)	Explain the Coefficient of Determination (R^2) and how it measures model performance.	(Understand)	6
B)	What is multivariable regression, and how does it extend linear regression to multiple features?	(Remember)	6
C)	Demonstrate how to implement logistic regression using the scikit-learn library.	(Evaluation)	6
	*** End ***		